

1 Set theory

1.4 Axiom of Choice

The most commonly used axiomatic system for mathematics is the Zermelo-Fraenkel set theory (ZF).

- ZFC with the Axiom of Choice
- ZF without the Axiom of Choice

Let's see what is this Axiom of Choice.

(Axiom of Choice) If $\{A_i\}_{i \in I}$ is a family of nonempty pairwise disjoint sets, there is a set $B \subset \bigcup A_i$ such that $B \cap A_i$ is exactly one element for all $i \in I$.

What if the sets are not disjoint?

Definition 1.44. Let $\mathcal{A} = \{A_i\}_{i \in I}$ be a family of nonempty sets. A choice function for $\mathcal{A}$ is a function $f : \mathcal{A} \rightarrow \bigcup_{i \in I} A_i$ such that $f(i) \in A_i$ for all $i \in I$.

Proposition 1.45. *For any collection of nonempty sets, there exists a choice function.*

Proof. Let $\mathcal{A} = \{A_i\}_{i \in I}$ be a family of nonempty sets. For each $i \in I$ set

$$B_i := \{(i, a) : a \in A_i\}.$$

Then the family $\{B_i\}_{i \in I}$ consists of nonempty pairwise disjoint sets. By the Axiom of Choice (the pairwise disjoint formulation) there exists a set $C \subseteq \bigcup_{i \in I} B_i$ such that $C \cap B_i$ contains exactly one element for every $i \in I$. For each i write the unique element of $C \cap B_i$ as (i, c_i) with $c_i \in A_i$, and define $f : \mathcal{A} \rightarrow \bigcup_{i \in I} A_i$ by $f(A_i) = c_i$. Equivalently, view f as a function $f : I \rightarrow \bigcup_{i \in I} A_i$ with $f(i) = c_i$. In either view $f(i) \in A_i$ for all i , so f is a choice function for $\mathcal{A}$. $\square$

The reason AC is debatable is that this doesn't constructively provide a way to choose elements from each set. If you have a finite collection of sets, you can simply list the elements and choose one from each set. But when you have an infinite collection of sets, the Axiom of Choice becomes more controversial.

1.4.1 Maximum principle and Zorn's lemma

Definition 1.46. A chain is a totally (linearly) ordered subset of a poset.

Definition 1.47. Let A be a poset and $B \subseteq A$. Then, an upper bound of B in A is an element $u \in A$ such that $b \leq u$ for all $b \in B$.

Example 1.48. Consider the set $A = \{1, 2, 3\}$ and the subset $\{\emptyset, \{2\}, \{1, 2, 3\}\} \subset P(A)$ is a chain.

(Maximum Principle) In a poset, every chain is contained in a maximal chain (i.e., a chain that is not properly contained in any other chain).

(Zorn's Lemma) Let A be a poset. If every chain in A has an upper bound in A , then A has a maximal element.

Proposition 1.49 (Maximum principle implies Zorn's lemma). *If we assume the Maximal Principle is true for all posets, then Zorn's Lemma is true for all posets.*

Proof. Let A be a poset satisfying the assumptions of Zorn's lemma. Let $B \subset A$ be a maximal chain. Let u be an upper bound of B in A . If $u < v$, then $B \cup \{v\}$ is a chain containing B , contradicting the maximality of B . Thus, u is a maximal element of A . $\square$

1.4.2 Four equivalent statements

Theorem 1.50. *The following statements are equivalent:*

1. (Axiom of Choice) *If $\{A_i\}_{i \in I}$ is a family of nonempty pairwise disjoint sets, there is a set $B \subset \bigcup A_i$ such that $B \cap A_i$ is exactly one element for all $i \in I$.*
2. (Zorn's Lemma) *If each chain (linearly ordered set) in a nonempty poset A has an upper bound, then A has a maximal element.*
3. (Maximal Principle) *In a poset, every chain is contained in a maximal chain.*
4. (Zermelo's Well-Ordering Theorem) *Every set can be well-ordered.*

Proof. We've already seen how 3 implies 2. Let's see one more.

$4 \Rightarrow 1$ Make $U := \bigcup A_i$ well ordered. Then, pick $a_i := \min A_i$ since $A_i \subset U$. Then, $B = \{a_i\}_{i \in I}$ is a set such that $B \cap A_i$ is exactly one element for all $i \in I$. Thus, the Axiom of Choice holds.

$2 \Rightarrow 1$. Assume Zorn's Lemma. Let $\{A_i\}_{i \in I}$ be a family of nonempty (pairwise disjoint) sets. Consider the set

$$\mathcal{F} = \{f \mid f \text{ is a function with domain } J \subseteq I, f(j) \in A_j \forall j \in J\},$$

partially ordered by extension ($f \leq g$ iff $f \subseteq g$). If $\mathcal{C} \subseteq \mathcal{F}$ is a chain, then $g := \bigcup_{f \in \mathcal{C}} f$ is an upper bound of $\mathcal{C}$ (the union is a function since members of the chain agree on overlaps). By Zorn's Lemma there exists a maximal element $f^* \in \mathcal{F}$. If $\text{dom } f^* \neq I$ pick $i_0 \in I \setminus \text{dom } f^*$ and choose $a \in A_{i_0}$ (nonemptiness). Then $f^* \cup \{(i_0, a)\} \in \mathcal{F}$ strictly extends f^* , contradicting maximality. Hence $\text{dom } f^* = I$, so f^* is a choice function. $\square$

1.5 Cardinality

For any two sets A and B , we say they have the same *cardinality* if there exists a bijection $f: A \rightarrow B$. Intuitively, this means that the sets “have the same number of elements”.

Having the same cardinality is an equivalence relation on the class of all sets.

We write $|A| = |B|$ to denote that the sets A and B have the same cardinality.

We say A has cardinality less than or equal to B (or B has cardinality greater than or equal to A), denoted $|A| \leq |B|$ or $|B| \geq |A|$, if there exists an injection $f: A \rightarrow B$.

Remark 1.51. By convention, we say the emptyset $|\emptyset| \leq |A|$ for any set A .

Lemma 1.52. *Consider two nonempty sets A and B . Then, $|A| \leq |B|$ if and only if there exists a surjective function $g: B \rightarrow A$.*

Proof. ($\Rightarrow$) Suppose there is an injection $f: A \rightarrow B$. Fix $a_0 \in A$. Define $g: B \rightarrow A$ by

$$g(b) = \begin{cases} f^{-1}(b), & b \in f(A), \\ a_0, & b \notin f(A). \end{cases}$$

Then g is surjective.

($\Leftarrow$) Suppose there is a surjection $g: B \rightarrow A$. For each $a \in A$ choose (one) $b_a \in g^{-1}(a)$; this choice defines a function $h: A \rightarrow B$, $h(a) = b_a$. Now for any $a \neq a' \in A$, we have that $g^{-1}(a) \cap g^{-1}(a') = \emptyset$, so $h(a) \neq h(a')$. Thus, h is injective. $\square$

You might ask whether we can prove the second direction without AC. Unfortunately no as they are equivalent. Think about it.

Theorem 1.53. 1. $|A| \leq |A|$

2. $|A| \leq |B|$ and $|B| \leq |C|$ imply $|A| \leq |C|$

3. If $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$ (This is quite nontrivial! It is the Bernstein-Cantor-Schröder theorem.)

4. For any two sets A, B , either $|A| \leq |B|$ or $|B| \leq |A|$.

Proof. the first two are trivial. The third one requires a bit more work and we will leave it to the end of this part. We can use the well-ordering theorem to prove 4 here.

For any two sets A, B , we first well-order them to be $(A, \leq_A)$ and $(B, \leq_B)$ using the well-ordering theorem. Then, by Theorem 1.43, we have that one poset is order-isomorphic to an initial segment of the other. This isomorphism gives rise to an injective map that we want and hence concludes the proof. $\square$

1.5.1 Finite sets

Now, let's consider the finite case.

Definition 1.54. A set A is said to be *finite* if there is a bijection $f: A \rightarrow \{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$ or A is empty. Otherwise, we say that A is *infinite*.

We intuitively say that A has cardinality n or 0.

Goal: Show that cardinality of a finite set is unique.

Lemma 1.55. Let A be finite and $a_0 \in A$. Then, there exists a bijection $f: A \rightarrow \{1, \dots, n+1\}$ for some $n \in \mathbb{N}$ if and only if there exists a bijection $g: A \setminus \{a_0\} \rightarrow \{1, \dots, n\}$ for some $n \in \mathbb{N}$.

Proof. ($\Rightarrow$) Suppose that $f: A \rightarrow \{1, \dots, n+1\}$ is a bijection, and let $k = f(a_0)$. The restriction of f is a bijection

$$f|_{A \setminus \{a_0\}}: A \setminus \{a_0\} \longrightarrow \{1, \dots, n+1\} \setminus \{k\}.$$

Define $h: \{1, \dots, n+1\} \setminus \{k\} \rightarrow \{1, \dots, n\}$ by

$$h(j) = \begin{cases} j, & j < k, \\ j-1, & j > k. \end{cases}$$

Then h is a bijection, so $g = h \circ f|_{A \setminus \{a_0\}}$ is a bijection from $A \setminus \{a_0\}$ to $\{1, \dots, n\}$.

($\Leftarrow$) Conversely, suppose that $g: A \setminus \{a_0\} \rightarrow \{1, \dots, n\}$ is a bijection. Define $f: A \rightarrow \{1, \dots, n+1\}$ by

$$f(a) = \begin{cases} g(a), & a \neq a_0, \\ n+1, & a = a_0. \end{cases}$$

The restriction of f to $A \setminus \{a_0\}$ maps bijectively onto $\{1, \dots, n\}$, while a_0 maps to the remaining element $n+1$. Hence f is a bijection. $\square$

Theorem 1.56. Suppose $f: A \rightarrow \{1, \dots, n\}$ is a bijection and let $B \subsetneq A$. Then, there is no bijection $g: B \rightarrow \{1, \dots, n\}$.

Proof. For $n = 1$, then $B = \emptyset$, and there is no bijection $g: \emptyset \rightarrow \{1\}$ (there is no even functions).

If the theorem is true for n , we will next show that it is true for $n+1$. Then, we use the inductive principle to conclude that the theorem is true for all $n \in \mathbb{N}$ (see homework).

Let $f: A \rightarrow \{1, \dots, n+1\}$ be a bijection. Let $B \subsetneq A$.

If $B = \emptyset$, same as before.

If $B \neq \emptyset$, choose $a_0 \in B$. Apply Theorem 1.55 to get a bijection $h: A \setminus \{a_0\} \rightarrow \{1, \dots, n\}$. Note $B \setminus \{a_0\} \subsetneq A \setminus \{a_0\}$. Since the theorem holds true for n , there is no bijection $g: B \setminus \{a_0\} \rightarrow \{1, \dots, n\}$. By Theorem 1.55 again, there is no bijection $g: B \rightarrow \{1, \dots, n+1\}$. $\square$

Corollary 1.57. *If A is finite, then there is no bijection from A onto a proper subset of itself.*

Proof. Suppose for contradiction that $f : A \rightarrow B$ is a bijection with $B \subsetneq A$. Let $g : A \rightarrow \{1, \dots, n\}$ be a bijection (existence follows from finiteness). Then the composite

$$g \circ f^{-1} : B \longrightarrow \{1, \dots, n\}$$

is a bijection from the proper subset B to $\{1, \dots, n\}$, contradicting Theorem 1.56. A commutative diagram summarising the argument:

$$\begin{array}{ccc} A & \xrightarrow[\cong]{f} & B \\ g \downarrow \cong & \swarrow g \circ f^{-1} & \\ \{1, \dots, n\} & & \end{array}$$

Hence such an f cannot exist. □

Corollary 1.58 (Uniqueness of finite cardinality). *If A is finite and there are bijections $f : A \rightarrow \{1, \dots, n\}$ and $g : A \rightarrow \{1, \dots, m\}$, then $n = m$.*

Proof. Suppose $m < n$. Then $g^{-1} : \{1, \dots, m\} \rightarrow A$ and $f : A \rightarrow \{1, \dots, n\}$ yield a bijection

$$f \circ g^{-1} : \{1, \dots, m\} \longrightarrow \{1, \dots, n\}.$$

Since $m < n$, the set $\{1, \dots, m\}$ is a proper subset of $\{1, \dots, n\}$, contradicting Theorem 1.57 applied to $A = \{1, \dots, n\}$. Similarly $n < m$ leads to a contradiction. Thus $n = m$. Diagrammatically:

$$\begin{array}{ccc} A & \xrightarrow[\cong]{f} & \{1, \dots, n\} \\ g \downarrow \cong & \nearrow f \circ g^{-1} & \\ \{1, \dots, m\} & & \end{array}$$

□

From now on, for a finite A with a bijection to $\{1, \dots, n\}$, we can say that the cardinality of A is n . So in the finite case, the cardinality is no different from counting the number of elements.

1.5.2 Infinite sets

Corollary 1.59. *The set $\mathbb{N} = \{1, 2, 3, \dots\}$ is infinite.*

Proof. Consider the map $f : \mathbb{N} \rightarrow \mathbb{N} \setminus \{1\}$ defined by $f(n) = n + 1$. This is a bijection onto the proper subset $\mathbb{N} \setminus \{1\}$. If $\mathbb{N}$ were finite, this would contradict Theorem 1.57. Hence $\mathbb{N}$ is not finite (i.e. infinite). □

Definition 1.60. A set A is

- *infinite* if it is not finite.
- *countably infinite* if it is in bijection with $\mathbb{N}$; in this case, we write $|A| = \aleph_0$.
- *countable* if it is either finite or countably infinite.
- *uncountable* if it is not countable.

Example 1.61. Now we try to compare $\mathbb{N}$ and $\mathbb{Z}$. We define a function $f: \mathbb{N} \rightarrow \mathbb{Z}$ as follows: we let $f(1) = 0$ and for $n \geq 0$,

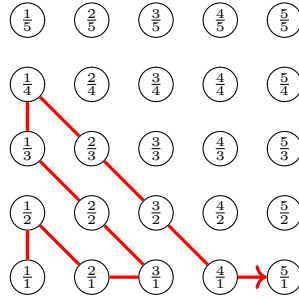
$$f(2n) = n, \quad f(2n+1) = -n.$$

The enumeration is visualised below:

$\mathbb{N}$:	1	2	3	4	5	6	7	8	$\dots$	$2i$	$2i+1$
	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$		$\downarrow$	$\downarrow$
$\mathbb{Z}$:	0	1	-1	2	-2	3	-3	4	$\dots$	i	$-i$

Hence f is a bijection, so $|\mathbb{Z}| = \aleph_0$.

Example 1.62. The set of positive rational numbers $\mathbb{Q}_+$ is countably infinite. We can enumerate it as follows: Represent each rational number as a pair $(m, n) \in \mathbb{N}^2$ by the (not necessarily reduced) fraction m/n . Enumerate lattice points by successive anti-diagonals in a zig-zag (Cantor) pattern:



Skipping repetitions (keep only reduced fractions) we obtain an enumeration of $\mathbb{Q}_+$, i.e., an bijection from $\mathbb{Q}_+$ to $\mathbb{N}$. Therefore, $|\mathbb{Q}_+| = \aleph_0$.

Now let's see some criteria for countability.

Lemma 1.63. Let $C \subset \mathbb{N}$ be infinite. Then C is countably infinite.

Proof. Define $f: \mathbb{N} \rightarrow C$ inductively as follows: let $f(1) = \min C$. If $f(1), \dots, f(n)$ have been defined, let $f(n+1) = \min(C \setminus \{f(1), \dots, f(n)\})$.

f is injective: for $m < n$, $f(n) \in C \setminus \{f(1), \dots, f(n-1)\}$, but $f(m)$ is not in the set. So, $f(n) \neq f(m)$.

f is surjective: for any $c \in C$, there exists $n \in \mathbb{N}$ such that $f(n) \geq c$. Let m be the smallest integer with $f(m) \geq c$. So for any $i < m$, $f(i) < c$. So $c \notin \{f(1), \dots, f(m-1)\}$ and hence $f(m) \leq c$ by its definition. This implies that $f(m) = c$. $\square$

The same proof can be used to prove the following result.

Proposition 1.64. *If A is infinite, then $|A| \geq \aleph_0$.*

Proof. Since A is infinite, so A is not empty and hence we can choose $\leq$ to make A well-ordered. We define a map $f : \mathbb{N} \rightarrow A$ inductive as follows: $f(1) = \min A$. If $f(1), \dots, f(n)$ have been defined, let $f(n+1) = \min(A \setminus \{f(1), \dots, f(n)\})$. Then, f is injective. So $|A| \geq \aleph_0$. $\square$

Theorem 1.65. *For any nonempty A , the following are equivalent:*

1. A is countable;
2. there exists a surjection $f : \mathbb{N} \rightarrow A$;
3. there exists an injection $g : A \rightarrow \mathbb{N}$.

Proof. We prove the implications as follows:

- (1) $\Rightarrow$ (2) If A is countably finite, then there exists a bijection and hence surjection $f : \mathbb{N} \rightarrow A$. If A is finite, then the composition is what we need: $\mathbb{N} \rightarrow \{1, \dots, n\} \rightarrow A$.
- (2) $\Rightarrow$ (3) If there exists a surjection $f : \mathbb{N} \rightarrow A$, then we can define an injection $g : A \rightarrow \mathbb{N}$ by $g(a) = \min\{n \in \mathbb{N} : f(n) = a\}$ for all $a \in A$.
- (3) $\Rightarrow$ (1) If there exists an injection $g : A \rightarrow \mathbb{N}$, then we can decompose this as $A \rightarrow \text{im}(g) \rightarrow \mathbb{N}$. If $\text{im}(g)$ is finite, then A is finite. If $\text{im}(g)$ is infinite, then by the lemma above, $\text{im}(g)$ is countably infinite and hence so is A . $\square$

Theorem 1.66. *The product of finitely many countable sets is countable.*

Instead of proving the general theorem, we show a special case.

Proposition 1.67. $|\mathbb{N} \times \mathbb{N}| = \aleph_0$.

Proof. We can define a function $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ by

$$f(m, n) = 2^m 3^n.$$

This function is injective, hence $\mathbb{N} \times \mathbb{N}$ is countable. Since $\mathbb{N} \times \mathbb{N}$ is obviously infinite, we have that $|\mathbb{N} \times \mathbb{N}| = \aleph_0$. $\square$